Module-X

Lecture-II and III

Scan Chain based Sequential Circuit Testing

1. Introduction

In the last lecture we have seen that the major problem in testing (and ATPG) of sequential circuits is difficulty in controlling secondary inputs (i.e., outputs of flip-flops) and difficulty in observing secondary outputs (i.e., inputs of flip-flops). In this lecture we will discuss various techniques to make the flip-flops controllable and observable, which convert a sequential circuit into virtual combinational one. Following that ATPG for combinational circuits would suffice for sequential circuits. However, for achieving this, additional circuitry, called design for test (DFT), will be put on-chip, which would add to extra area overhead.

In the next section we will explore different schemes to control and observe the flip-flops.

2. Controllability and observability of flip-flops

2.1. Set and reset lines

One of the simplest way to directly control flip-flops is through set-reset lines. Set-reset lines can directly make the output of a flip-flop to be 1/0 without any input and clock pulse. A D flip-flop with set-reset lines is shown in Figure 1 and its functionality is shown in Table 1.

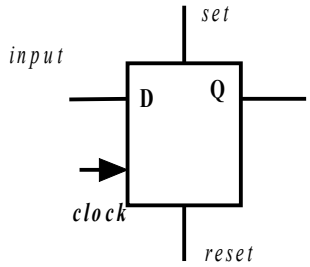


Figure 1. D-Flip-flop with set and reset

Table 1. Truth table for D flip-flop with set-reset lines

Input (D)	Output (Q)	*set*	*reset*	*clock*
Don't care	1	1	0	Don't care
Don't care	0	0	1	Don't care
Don't care	**Illegal**	1	1	Don't care
1	1	0	0	Clock edge
0	0	0	0	Clock edge

From the truth table it may be observed that *set (reset)* lines makes the output 1(0) irrespective of input D and clock pulse. However, both the set and reset lines cannot be made high simultaneously. Also, when the D-flip-flop functions normally, both set and reset lines are kept low.

Now we will see how use of this type of flip-flop helps in testing the circuit given in exercise of last lecture (Lecture 1, module 12). The circuit with a s-a-1 fault in the primary input line is shown in Figure 2 (a). Also, the flip-flop has set/reset lines. By D-algorithm, a test pattern would be: *primary input* =0, *secondary input* (other input of XOR gate)=0 and fault effect at *primary output*=**D.** So in Step-1 (Figure 2 (b)), *set*=1 and *reset*=0 (and and *primary input* =X); this makes output of flip-flop (i.e., secondary input) to be 1. In Step-2 (Figure 2 (c)), *set*=0, *reset*=0 and *primary input* =0; this sensitizes fault location as $\overline{D}$ and its effect propagates to the input of the flip-flop as **D**. Also a positive clock pulse is applied which transfers **D** to output of the flip-flop (primary output). These two steps complete ATPG (and testing) of the fault.

So it may be observed that the fault which was un-testable by time frame expansion method becomes testable using set/reset flip-flop. Further, one pattern is required to set/reset the flip-flops and another pattern (at primary inputs) is required to sensitize and propagate the fault effect to a primary output. So, unlike time frame expansion method where d_{seq}+1 test patterns are required (d_{seq} patterns to initialize the flip-flops and one pattern to sensitize/propagate fault effects), in case of set/reset flip-flops only two patters are required (one to set or reset the flops and one to sensitize/propagate fault effects). This saving in number of test patterns (i.e., test time) is illustrated in the circuit example of Figure 10 of the last lecture. The circuit (Figure 10 of Lecture 1, module 12) with set/reset flip flops is shown in Figure 3(a).

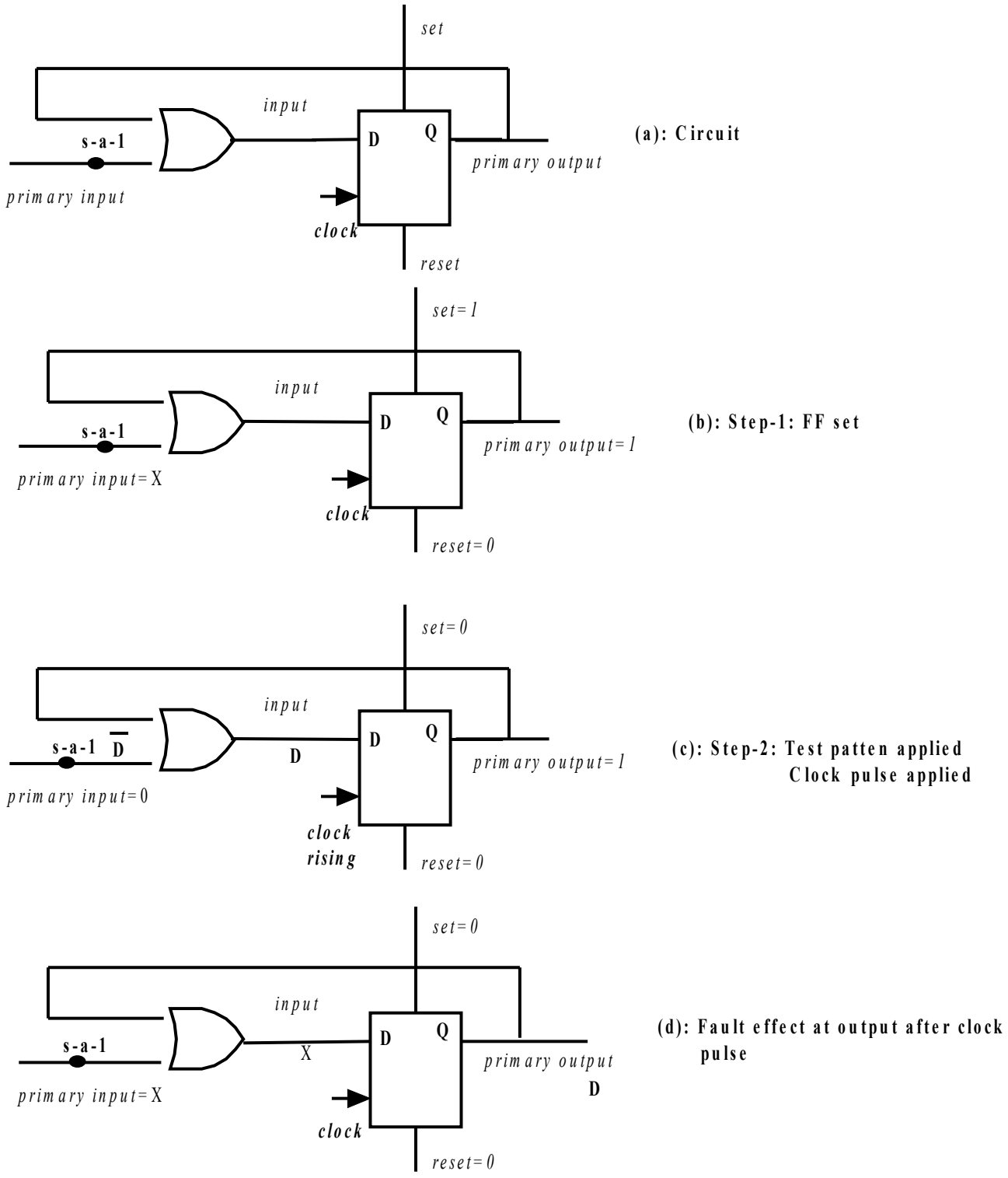


Figure 2. Testing using set/reset flip-flops for a cyclic circuit

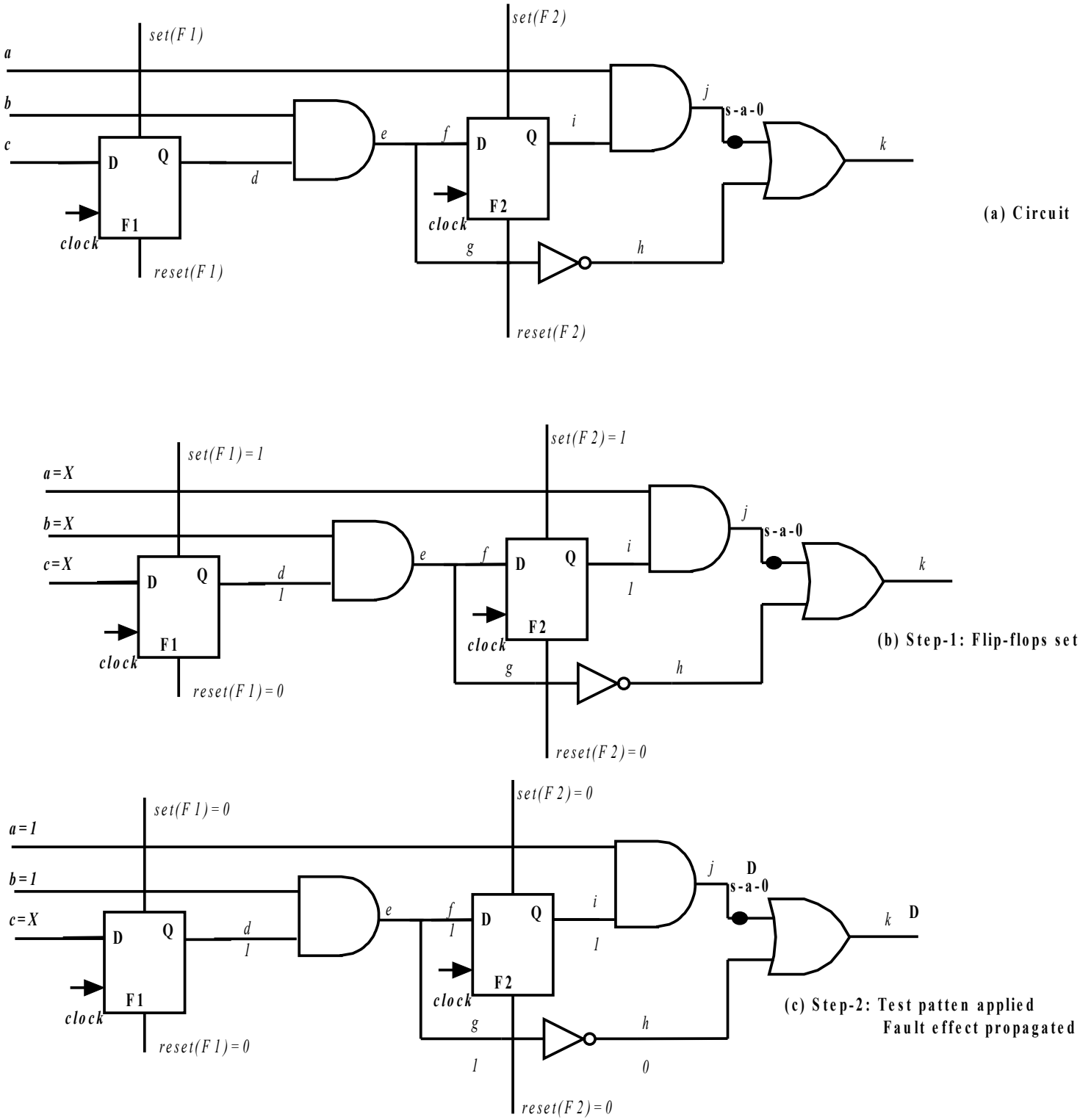


Figure 3. Testing using set/reset flip-flops: illustration of saving in test patterns

As discussed in the last lecture, three patterns were required to test the s-a-0 fault at net *j:* (i) *a*=X,*b*=X,*c*=1 and clock pulse (ii) *a*=X,*b*=1,*c*=1 and clock pulse and (iii) *a*=1,*b*=1,*c*=X. Also it is to be noted that d_{seq}=2, for the circuit.

Now, by using the set/reset flip-flops, we will see only two patterns are required to test the fault. By ATPG using D-algorithm, nets *d* and *i* are to be 1 for testing the fault. So both the flip-flops are set by the pattern: *a*=X,*b*=X,*c*=X, *set*(F1)=1, *reset*(F1)=0, *set*(F2)=1 and *reset*(F2)=0; this is shown in Figure 3(b). Finally, pattern *a*=1,*b*=1,*c*=X, *set*(F1)=0, *reset*(F1)=0, *set*(F2)=0 and *reset*(F2)=0 is applied to sensitize and propagate the effect of the fault to primary output; this is shown in Figure 3(c).

So, only two patterns can test the fault. However the most important point is **"Irrespective of the value of d_{seq}, only two patterns are required to test a sequential circuit with set/reset flip-flops".** The gain can be easily understood because any practical circuit has more than thousands of flip-flops. On the other hand, there is a big problem with the scheme that makes it impractical to be applied in a practical system. An input output block diagram of the circuit of Figure 3 is shown in Figure 4. It may be noted that it has 9 I/O pins (3 primary inputs + 1 primary output + $2 \times n_{ffs}$ set-reset lines, where n_{ffs} is the number of flip-flops and a clock). The largest number of I/O pins supported in most complicated packages is about 1024. So, for a circuit with thousands of flip-flops, this approach requires a package of thousands of I/O pins (for the $2 \times n_{ffs}$ factor) with makes it impractical.

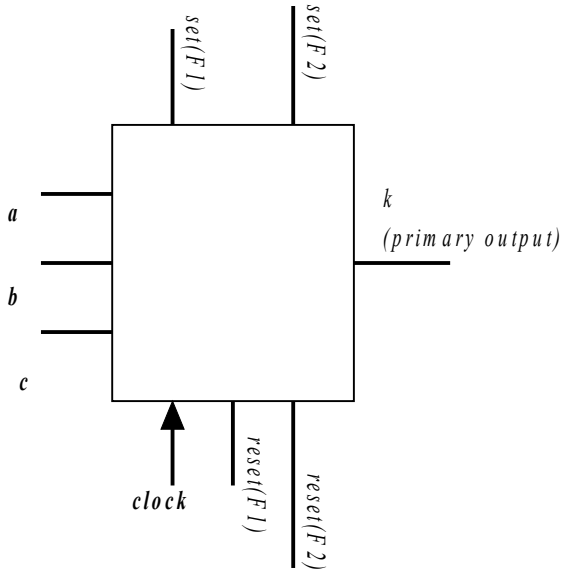
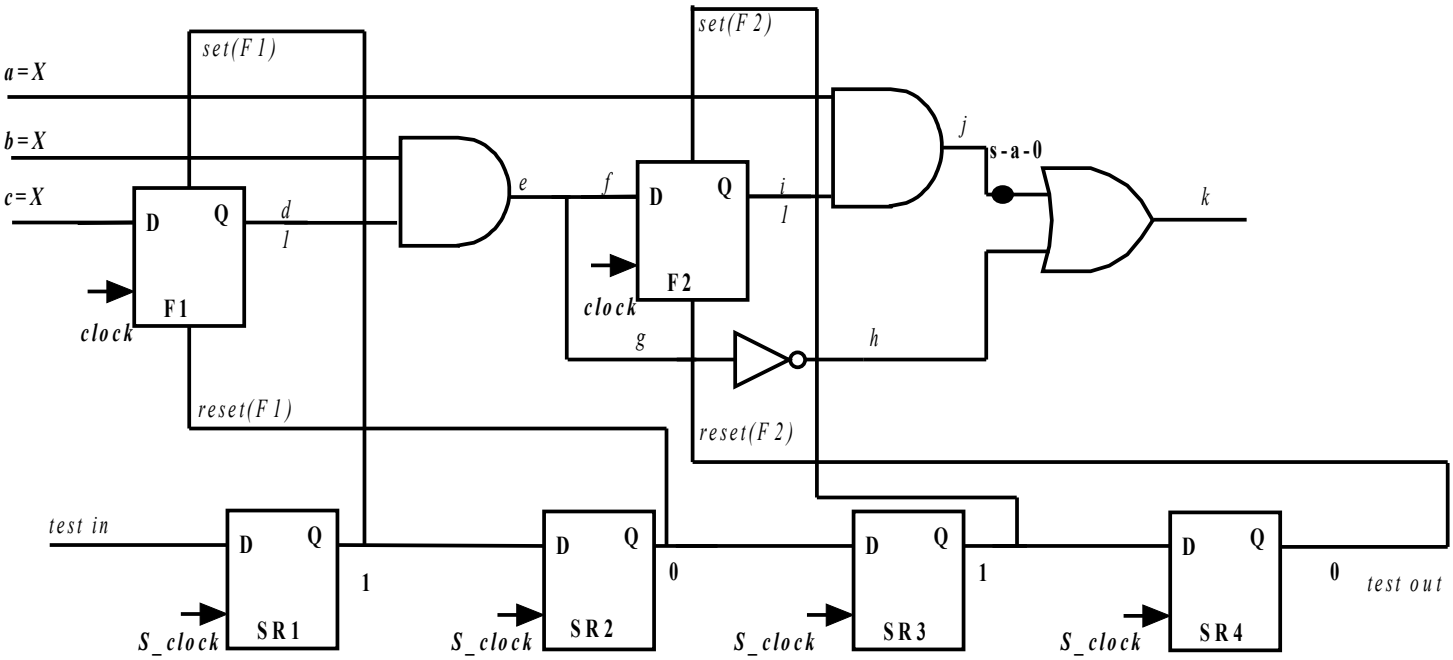


Figure 4. Input output block diagram of the circuit shown in Figure 3.

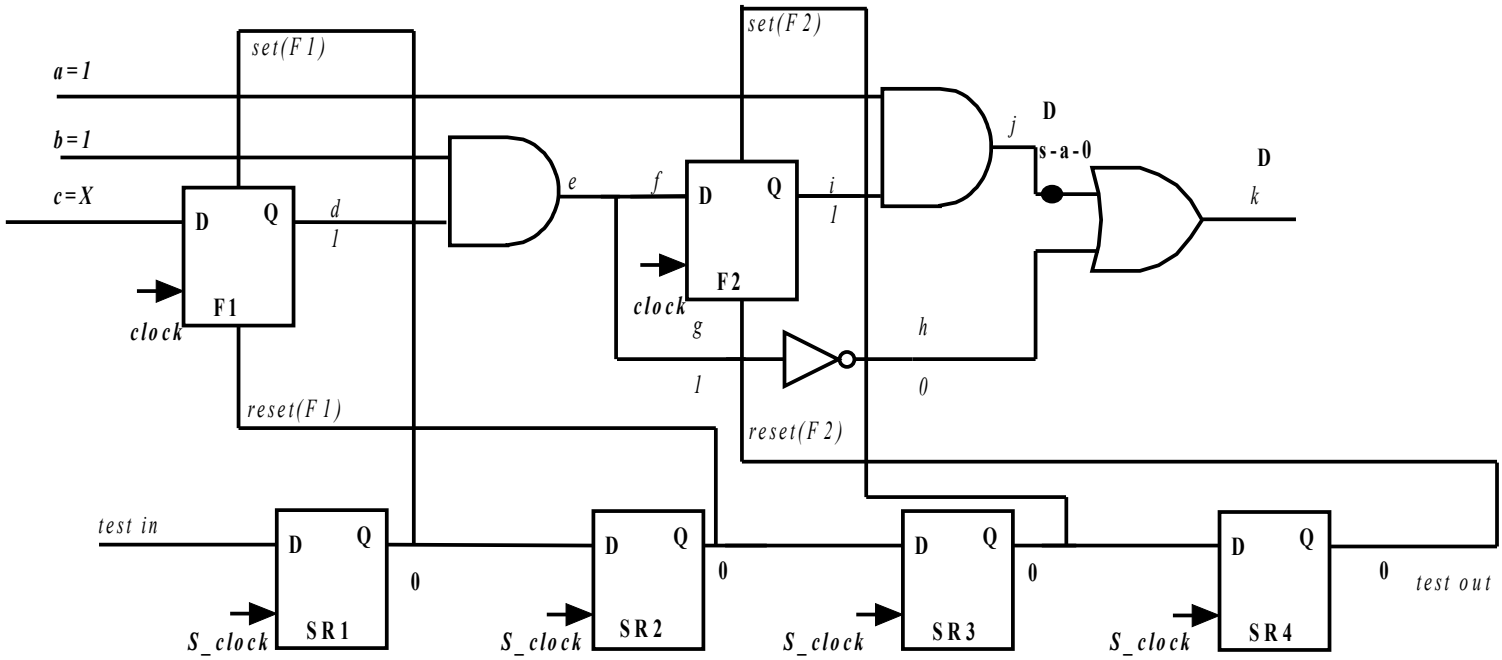
2.2 **Set and reset by shift register**

In the last sub-section we saw that flip-flops with set/reset capability can ease the testing and ATPG problem for sequential circuits. However, the main problem is the huge number of I/O lines that gets added to the chip. To avoid this issue of high I/Os, another technique called "set and reset by shift register" is used which uses a shift register and loads itself with the pattern required for setting the flip-flops. Now the set and reset lines of the flip-flops of the circuit (under test) are connected with the outputs of the flip-flops of the shift resister. The shift register has an input line and a separate clock (for shifting data) and an output line. So in this case only three extra I/O lines are required. ATPG and testing for the circuit shown in Figure 3, using shift register is illustrated in Figure 5.

As there are two flip-flops in the circuit under test, there are four flip-flops in the shift register. Now, set and reset lines of F1 (flip-flop) of the circuit under test are connected to output of SR1 and SR2 (flip-flops of register), respectively. Similarly, set and reset lines of F2 are connected to output of SR3 and SR4, respectively. As already discussed, testing the fault (s-a-0 at *j*) requires the two flip-flops of the circuit to be set to 1. So the pattern is *set*(F1)=1, *reset*(F1)=0, *set*(F2)=1 and *reset*(F2)=0, which when mapped to the flip-flops of the shit register is SR1=1,SR2=0,SR3=1 and SR4=0. The pattern 1010 can be inserted in the shift register using four clock edges of *S_clock*. Once the shift register is loaded, the flip-flops F1 and F2 are set; this is shown in Figure 5(a). Now, another pattern 0000 is inserted in the shift register using four clock edges of *S_clock;* this makes the circuit under test ready to be operational. Following that input pattern *a*=1,*b*=1,*c*=X is applied to sensitize and propagate effect of the fault to primary output; this is shown in Figure 5(b). The block diagram of the circuit (under test) with shift register is shown in Figure 6. So in addition to original I/Os, three new I/Os get added namely, (i) *S_clock*, (ii) *test in* and (iii) *test out*.



(a) Step-1:Shifting patten 1001 in the shift register
1001 in *test in* @ 4 S_clock edges



(b) Step-2:Shifting patten 0000 in the shift register
0000 in *test in* @ 4 S_clock edges

Step-3: Apply test patten at input *a*=1,*b*=1,*c*=X

Figure 5. Testing using set/reset flip-flops and shift register

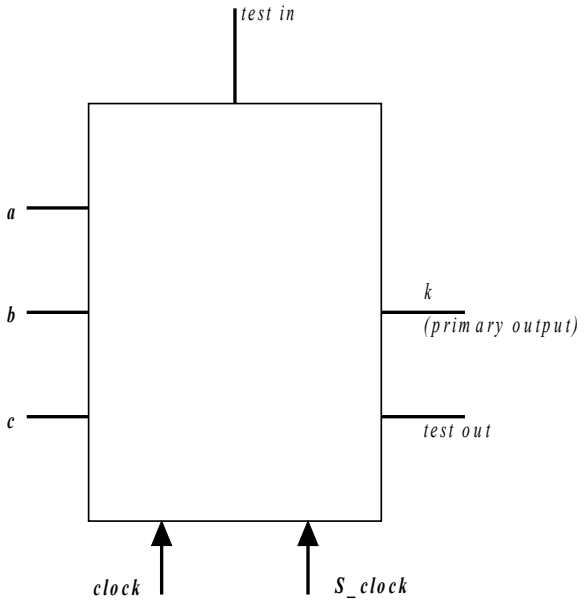


Figure 6. Input output block diagram of the circuit under test (Figure 5) with shift register

The following are the pros and cons of shift register based testing and ATPG.

Pros:

- ATPG algorithm does not require time frame expansion approach and flip-flops can be directly controlled from the register. So ATPG complexity is lower than time frame expansion approach.
- I/O requirements for testing is minimal (only 3)

Cons:

- Area overhead of the shift register is high (twice the number of flip-flops in the circuit)
- Test pattern application time is almost same as in the time frame expansion approach. Applying bit sequence in the shift register (required for setting/resetting the flip-flops) takes $2 \times n_{ffs}$ pulses of the clock of the register.

In the next section we will discuss another scheme similar to shift resister based testing called "scan chain". In scan chain based technique the flip-flops are directly controllable and I/O requirements are minimal. However, on the other side there is no requirement of the additional shift register and separate set/reset lines in flip-flops; due to these two points scan chain based scheme is widely accepted in VLSI testing.

3. **Scan Chain based Testing and ATPG for sequential circuits**

In the last section we saw that shift register based testing of sequential circuits solved the problem of a large number of I/Os. However, the major drawback of the scheme is due to the huge area overhead; twice the number of flip-flops in the circuit under test is required in the shift register. The concept of Scan chin is motivated from the idea of shift register based testing, however, alleviates the problem of area overhead [1]. The basic idea in scan chain is to convert the flip-flops in the circuit under test itself to a shift register, rather than using a separate one. To elaborate, the circuit under test has two modes: (i) test (or scan chain) and (ii) working (or normal). In test mode the flip-flops are decoupled from the circuit and they are connected in form of a shift register (called scan chain). The input of the first flip-flop in the scan chain is taken out as a special input test pin called "Scan in" and the output of the last flip-flop in the scan chain is taken out as a special output test pin called "Scan out". Now all the flip-flops are set as required by shifting bit values in the scan chain. Once the flip-flops are set they are removed from the scan chain and connected back to the circuit. So another advantage of scan chain based testing over shift register based testing is due to non requirement of set/reset lines. The basic concept of scan chain is shown in Figure 7.

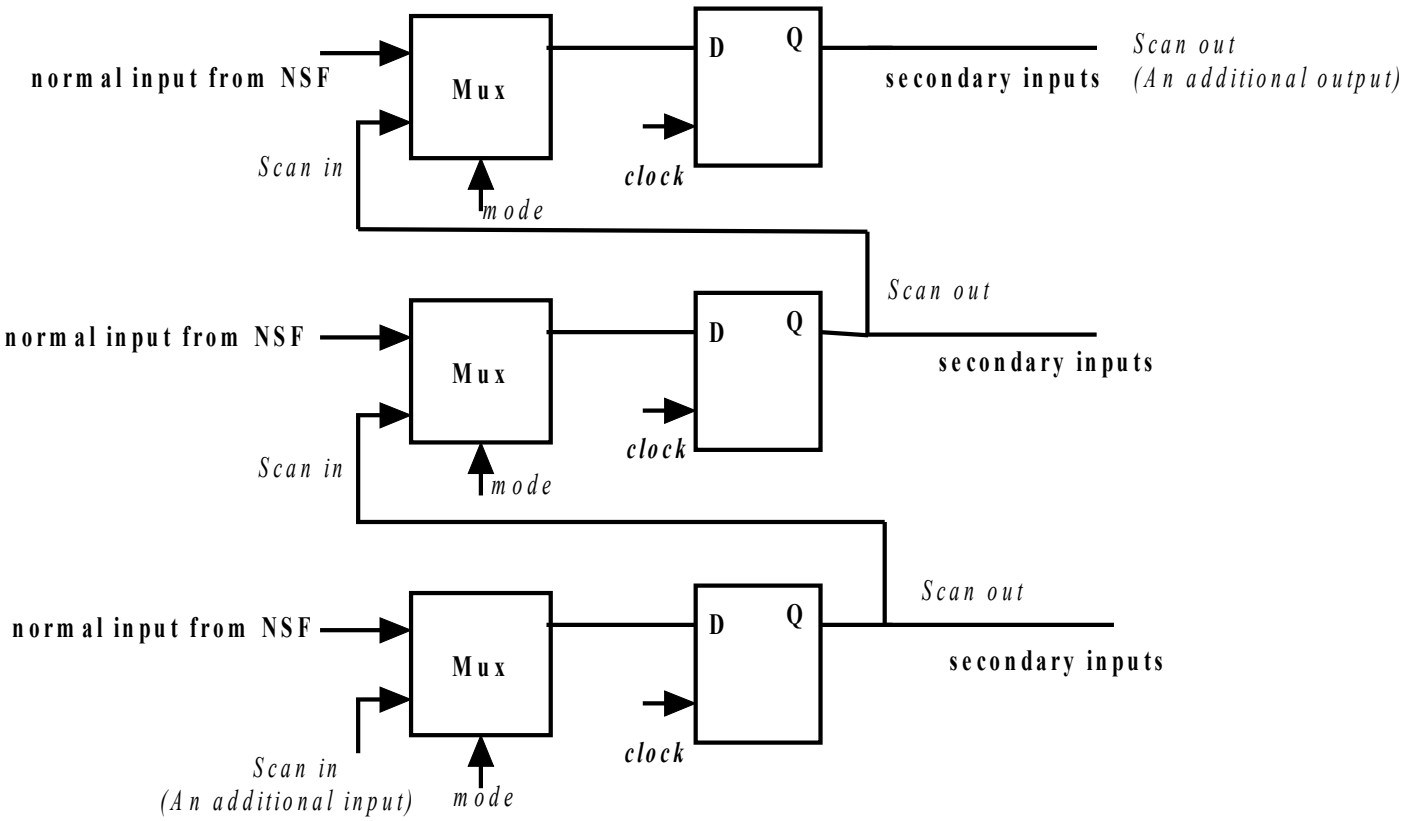


Figure 7. Scan chain in state flip-flop block

Figure 7 shows the state flip-flops with scan chain in a sequential circuit; other blocks are not shown for clarity. As discussed, flip-flops with scan-chain take inputs from two parts of the circuit (i) next state function block (when operating normally) and (ii) from the previous flip-flop in the chain (when in test mode); the first flip-flop in the scan chain takes input from a primary input "Scan in". So in state flip-flop block with scan chain, all the flip-flops must have a 2X1 multiplexer. One input is from the next state function block and the other is from the previous flip-flop in the chain. The control input called mode basically decides the mode of operation; 1 for scan-chain and 0 for normal operation. Figure 7 shows this concept for a circuit with three flip-flops. Figure 8 shows a flip-flop used in scan chain called **scan flip-flop** and Figure 9 shows the block diagram representation of scan flip-flop. Figure 10 shows the flip-flops of Figure 7 in scan chain mode and Figure 11 shows the flip-flops of Figure 7 in normal mode. Figure 12 shows the basic block diagram of a sequential circuit with scan chain in both normal and test mode.

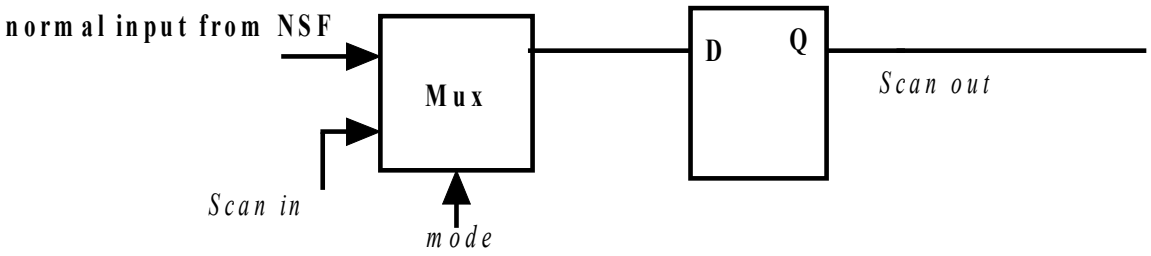


Figure 8. A scan flip-flop

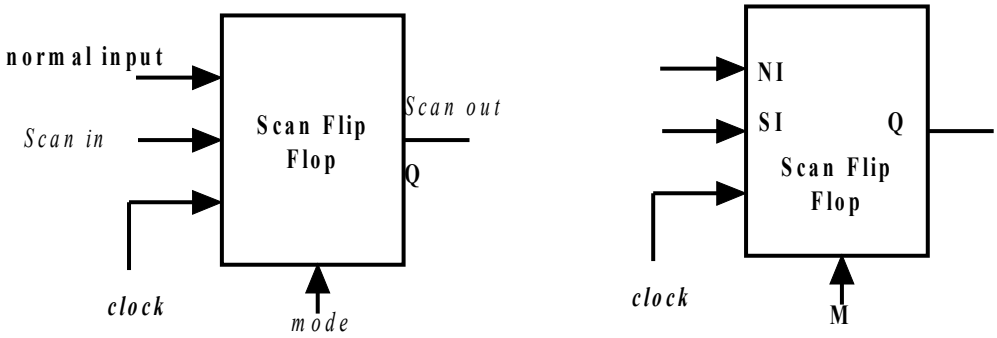


Figure 9. Block diagram of a scan flip-flop

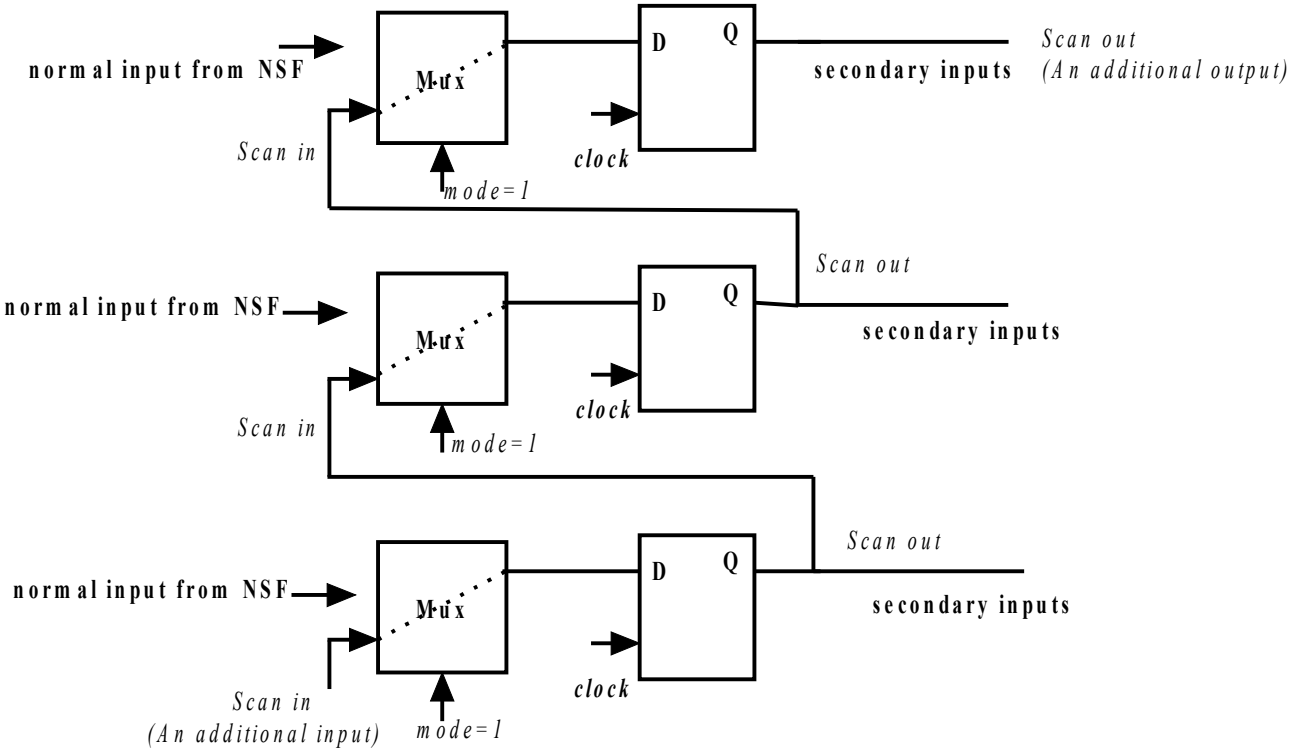


Figure 10. Flip-flops of Figure 7 in scan chain mode

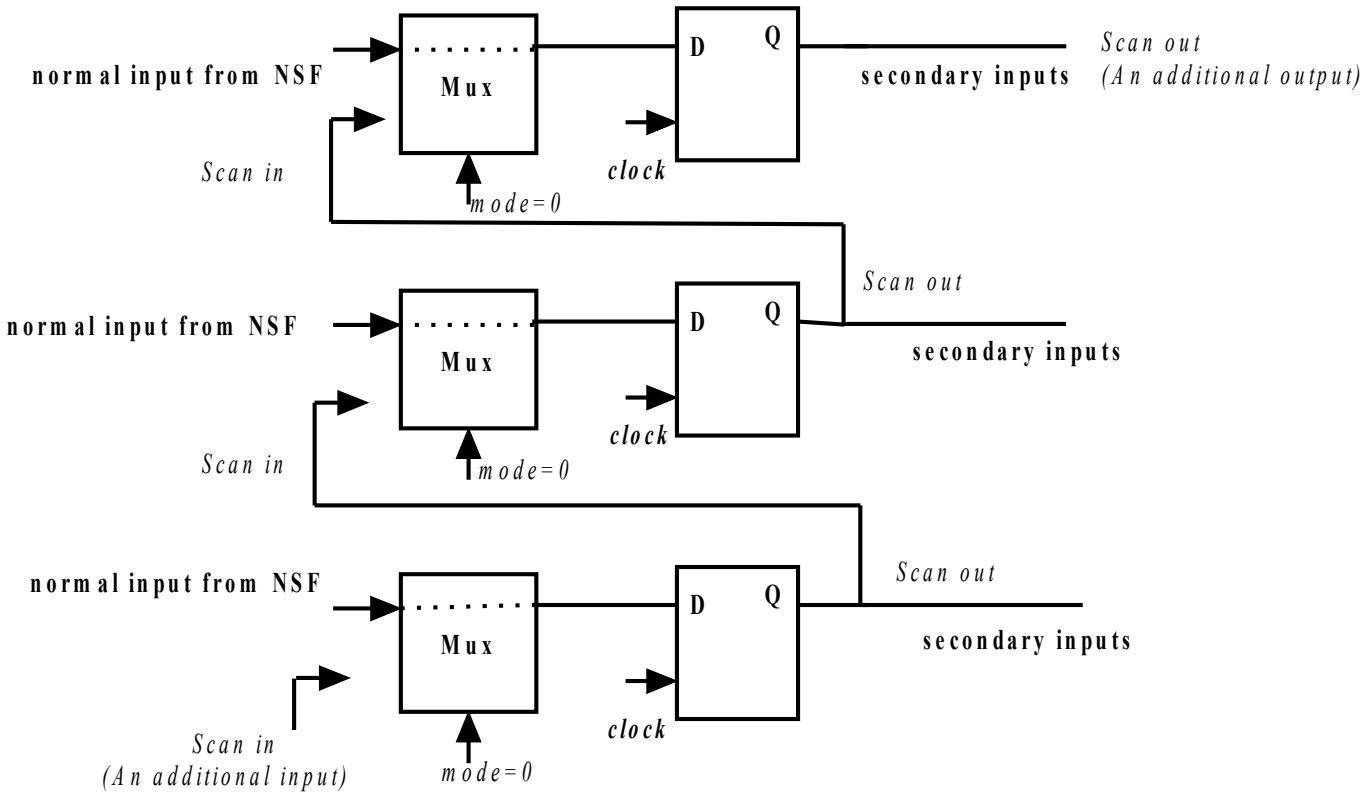
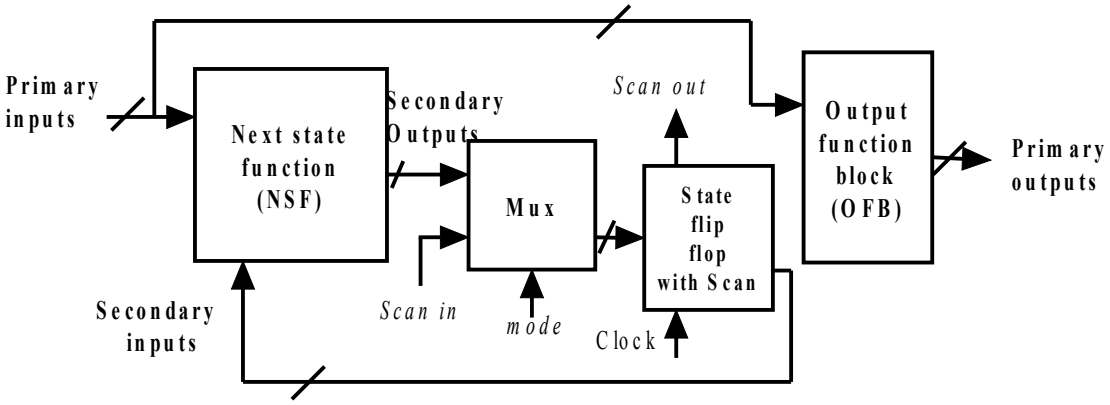
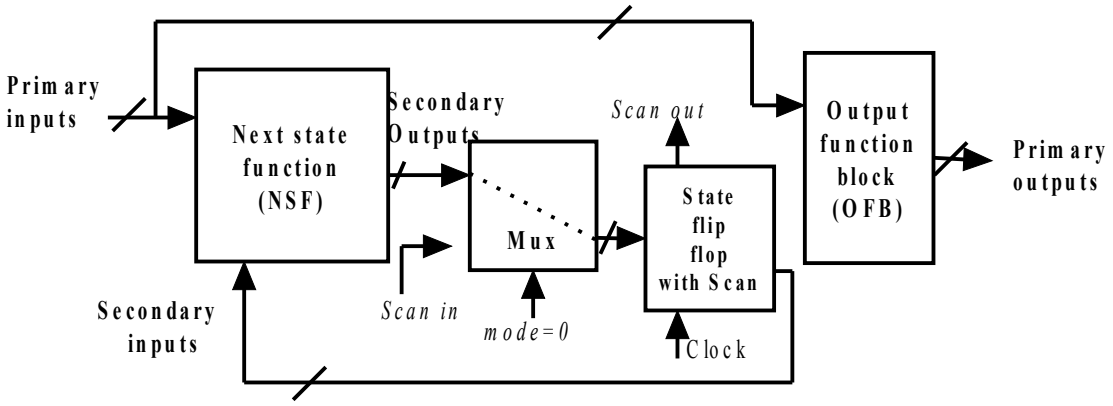


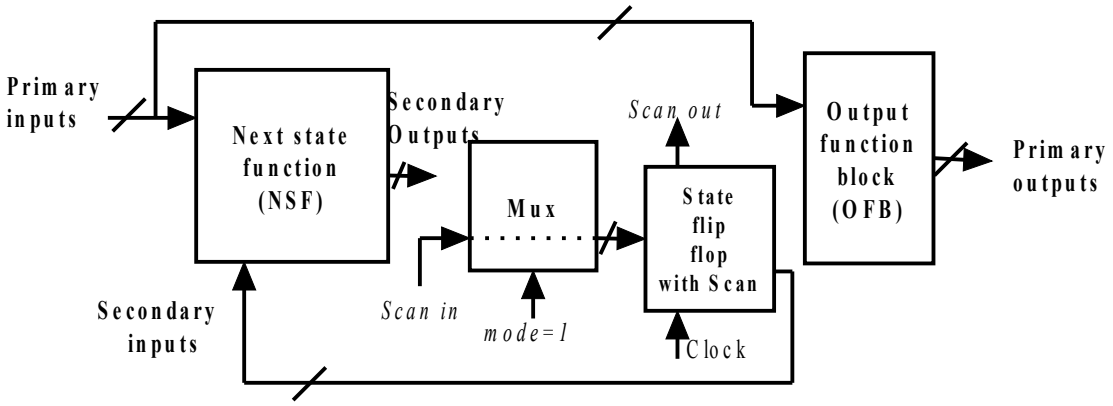
Figure 11. Flip-flops of Figure 7 in normal mode



(a) Sequential Circuit with Scan Chain



(b) Sequential Circuit with Scan Chain: Normal Mode



(c) Sequential Circuit with Scan Chain: Test Mode

Figure 12.Sequentail circuit with scan chain: normal and test mode

4. **ATPG and testing using scan chain in a sequential circuit: An Example**

In this section we will discuss ATPG and testing of the sequential circuit discussed in last lecture (Module 12, Lecture 1, Figure 10), now with scan chain. The circuit with the flip-flops having scan chain compatibility is shown in Figure 13(a). As discussed about the circuit (in Module 12, Lecture 1, Figure 10), to test the s-a-0 fault at net *j,* the signal values at nets *d* and *i* are to be 1. So both the flip-flops are to be set to 1; this was achieved by making the set input as 1 and reset input as 0 in case of testing using set/reset flip-flops. In case of scan chain, to set the flip-flops, in the first step mode (M) is made 1. Making M=1, removes the next state function block from the circuit and the flip-flops are connected in a chain; this is shown by dotted lines in Figure 13(b). Two 1s are applied in the Scan in input at two clock pulses which makes *d*=1 and *i*=1. Now, the circuit is brought in normal mode by making M=0. In this stage testing is performed by making *a*=1,*b*=1,*c*=X which propagates the fault effect to the output; this is shown in Figure 13(c).

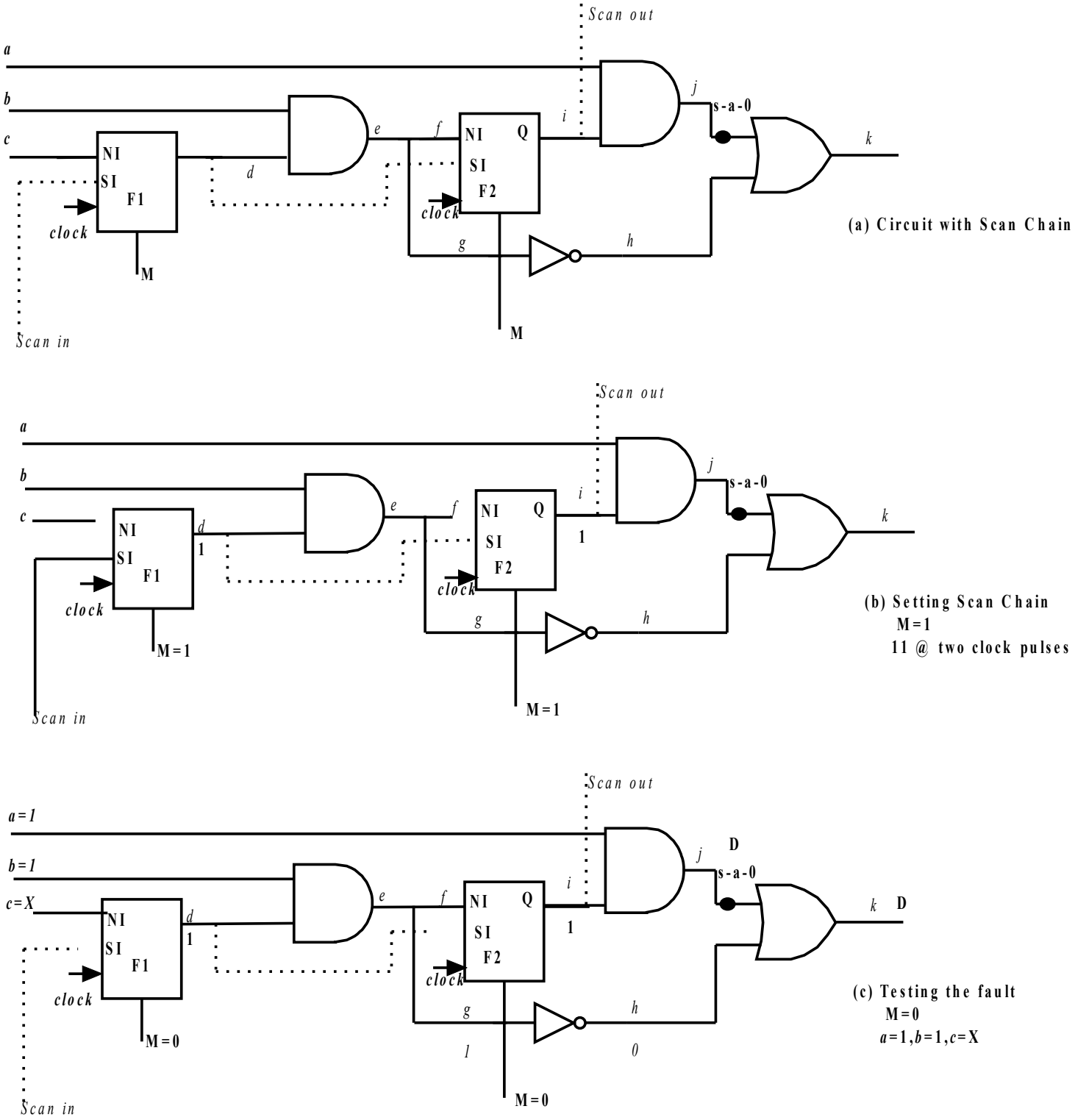


Figure 13. ATPG and testing of a sequential circuit (Figure 10, Module 12, Lecture 1) with scan chain

In the last example we saw how scan chain can be used to set the flip-flops to desired value for testing. However, D-algorithm is required on the entire circuit (after removing flip-flops) to find the test pattern. Now we will see another advantage of scan chain, which eliminates the need for performing ATPG on the whole circuit. One needs to consider input of the nearest flip-flop from the fault site as an output line and perform ATPG on the sub-circuit lying in the cone of influence of the output line (i.e., input of the nearest flip-flop). This concept will be illustrated on a circuit example given in Figure 14.

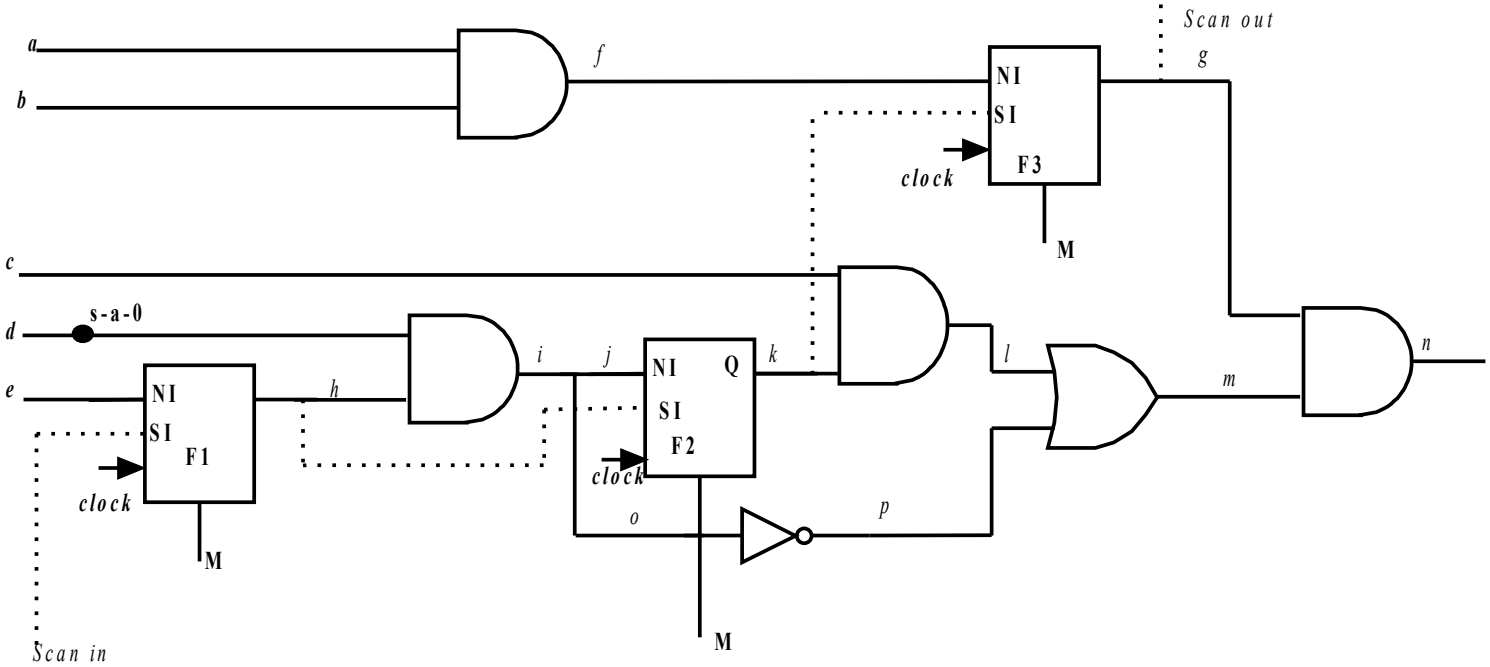


Figure 14. A sequential circuit with scan chain and s-a-0 fault at *d*

We will first illustrate the scheme of testing using scan chain where D-algorithm is applied on entire circuit and scan chain is used only to control the flip-flops; the procedure is shown in Figure 15. Figure 15 (a) illustrates D-algorithm applied on the full circuit of Figure 14 (after removing the flip-flops) where fault propagation path is taken as *d-i-o-p-m-n*. D-algorithm determines that *h* is to be 1 and *g* is to be 1; this implies that flip-flop F1 and F3 are to be set while F2 can be set or reset. Further, *a*=X,*b*=X,*c*=0,*d*=1,*e*=X would apply the pattern to sensitize and propagate the fault effect. So to make flip-flops F1 and F3 to be 1 and F2 to be X, pattern 1X1 is applied at Scan in input with three clock pulses keeping M=1; this is shown in Figure 15 (b). Finally, M is made 0 and test pattern *a*=X,*b*=X,*c*=0,*d*=1,*e*=X is applied, which propagates fault effect through primary output *n;* this is illustrated in Figure 15(c).

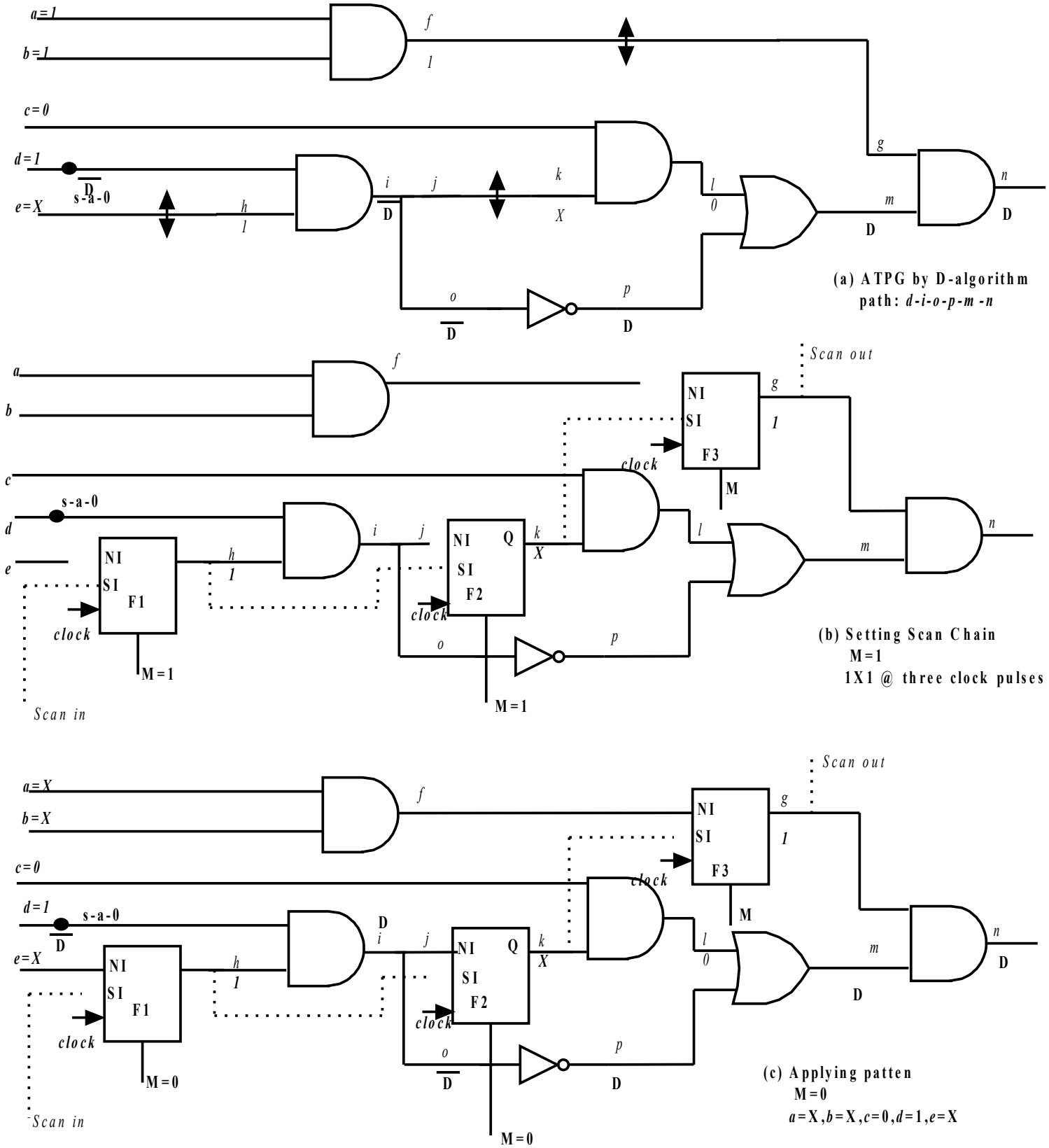


Figure 15. Testing using scan chain and fault propagation through circuit output

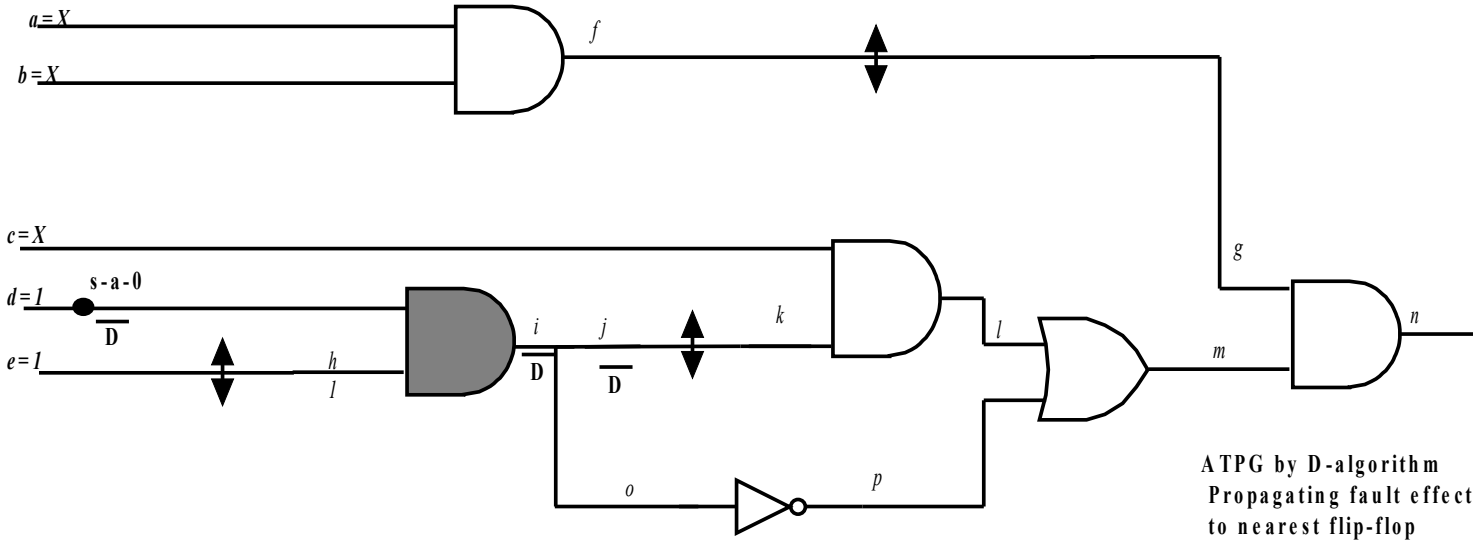


Figure 16. Fault sensitization and effect propagation to input of nearest flip-flop (of the circuit of Figure 14)

As already discussed, even in ATPG for sequential circuits, D-algorithm is performed on the full circuit with flip-flops removed. In the last example we saw the use of scan chain only to set/reset the flip-flops. Now we will see another use of scan chain in propagating the fault effect. So we need to just bring the fault effect to the input of the nearest flip-flop. This in turn requires ATPG on a sub-circuit whose output drives the input of the nearest flip-flop. In other words, ATPG is to be performed on the sub-circuit that lies in the cone of influence of the nearest flip-flop's input. Figure 16 illustrates that the nearest flip-flop from the fault site (s-a-0 at net *d,* of the circuit of Figure 14) is F2. So fault effect needs to be propagated to net *j* (which is input of F2). So, APTG is to be performed on the sub-circuit which lies in the cone of influence of net *j,* which includes inputs *d,e* and AND gate marked gray in Figure 16; test pattern is *d*=1,*e*=1 and fault effect $\overline{D}$ is propagated to net *j*. Obviously ATPG complexity is simplified. Now we will see how scan chain can propagate this fault effect out of the circuit. The whole testing procedure, when fault is propagated through scan chain (for the circuit of Figure 16) is illustrated in Figure 17 I and Figure 17 II.

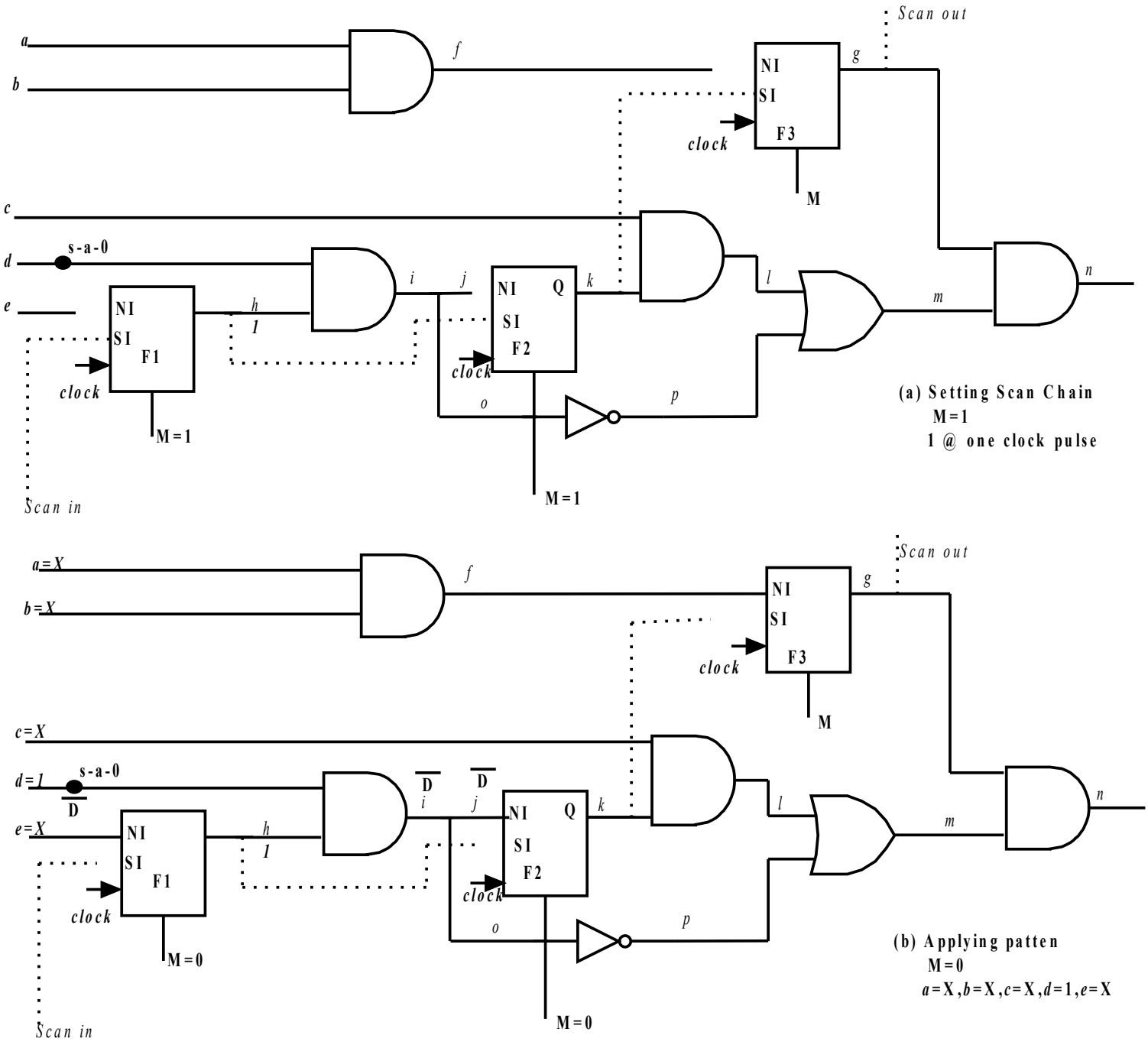


Figure 17-I. Fault effect propagation through scan chain (Step a and Step b for circuit of Figure 16)

In this case, as we are considering only a sub-circuit, so only F1 (i.e., net *h*) needs to be made 1; this is shown in Figure 16. This is achieved by making M=1, giving 1 in Scan in input and a clock pulse; shown in Figure 17-I(a). After that M is made 0 and pattern *d*=1,*e*=X is applied to sensitize the fault and propagate the effect ($\overline{D}$) to net *j* via net *i*; shown in Figure 17-I(b).

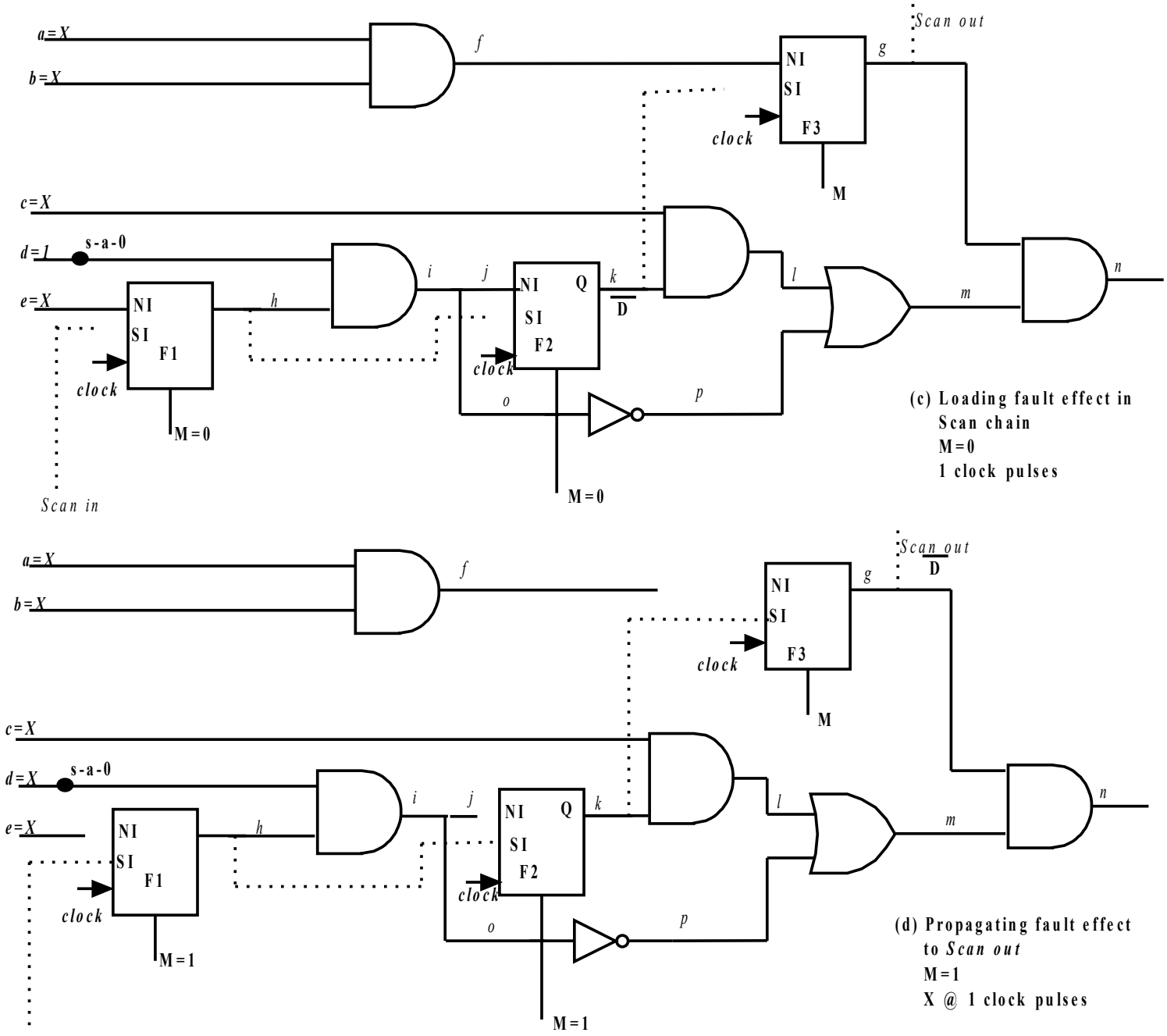


Figure 17-II. Fault effect propagation through scan chain (Step c and Step d for circuit of Figure 16)

Now the fault effect is to be brought out by the scan chain. This is achieved in two steps as shown in Figure 17-II. First, the effect $\overline{D}$ is loaded into the nearest flip-flop (F2) by applying a clock pulse keeping M=0; shown in Figure 17-II(a). Following that flip-flops are again brought into chain mode (by setting M=1) and 1 clock pulse is applied (with Scan in input as X), which brings the effect out through the Scan out output; shown in Figure 17-II(b).

5. **ATPG and testing using partial scan chain in a sequential circuit: An Example**

From the discussion in last lecture (Module 12, Lecture 1) we saw that a circuit with sequential dept d_{seq} needs d_{seq} clock pulses and patterns to set the flip-flops (to required values) when testing is done using time frame expansion approach. In case of scan chain if there are n_{ff} flip-flops, then n_{ff} clock pulses are required to set/reset the flip-flops. As $d_{seq} \leq n_{ff}$, test time is higher for scan chain based testing compared to time frame expansion method. Also, multiplexers are required in case of scan chains while no extra circuitry is required for time frame expansion method. Only ATPG complexity is lower in case of scan based testing. It may be noted that ATPG is off line exercise and test time is very expensive as patterns are applied by an automatic test equipment. However, scan based testing is still the most widely accepted technology. As discussed in the exercise of Lecture 1 Module 12, time frame expansion scheme cannot set/reset flip-flops which are cyclic (i.e., whose input is dependent on its own output). So scan based scheme or set/reset with shift register scheme is required for cyclic circuits.

A new scheme called "partial scan" came up taking ideas from both time frame expansion method and scan chain method [2]. The basic idea used in the scheme is to make scan enable only in those flip-flops which are cyclic and keep the remaining ones as normal. So we will have area overhead (because of MUX) only for those flip-flops which must be controlled directly and for the others, time frame expansion based indirect control can be used.

Circuit shown in Figure 18 will be used to illustrate the concept of partial scan.

Figure 18(a) shows a sequential circuit with D-algorithm applied after removing the flip-flops; the positions of the flip-flops are marked by double ended vertical arrows. It may be noted that two flip-flops are to be controlled to 1; net *d* is to be made 1 and net *i* is to be made 1. Making net *d*=1 is simple and can be achieved by applying *c*=1 and a clock pulse; this is shown in Figure 18(b). It may be noted that control of net *d* via F1 is by time frame expansion method; as d_{seq}=1 for F1, so one clock pulse and one pattern is enough to control it. To make net *i*=1, we need *b*=1,*d*=1,*m*=1 (which makes *f*=1 via net *e*) and a clock pulse. However there is a problem here. To make net *m*=1 we need *j*=1 which can be achieved by *a*=1,*i*=1. However to make *i*=1 we need *m*=1. This problem arises because flip-flop F2 is cyclic, as shown in Figure 18(b). So F2 cannot be controlled by time frame expansion method and it must have scan chain facility, as shown in Figure 18(c).

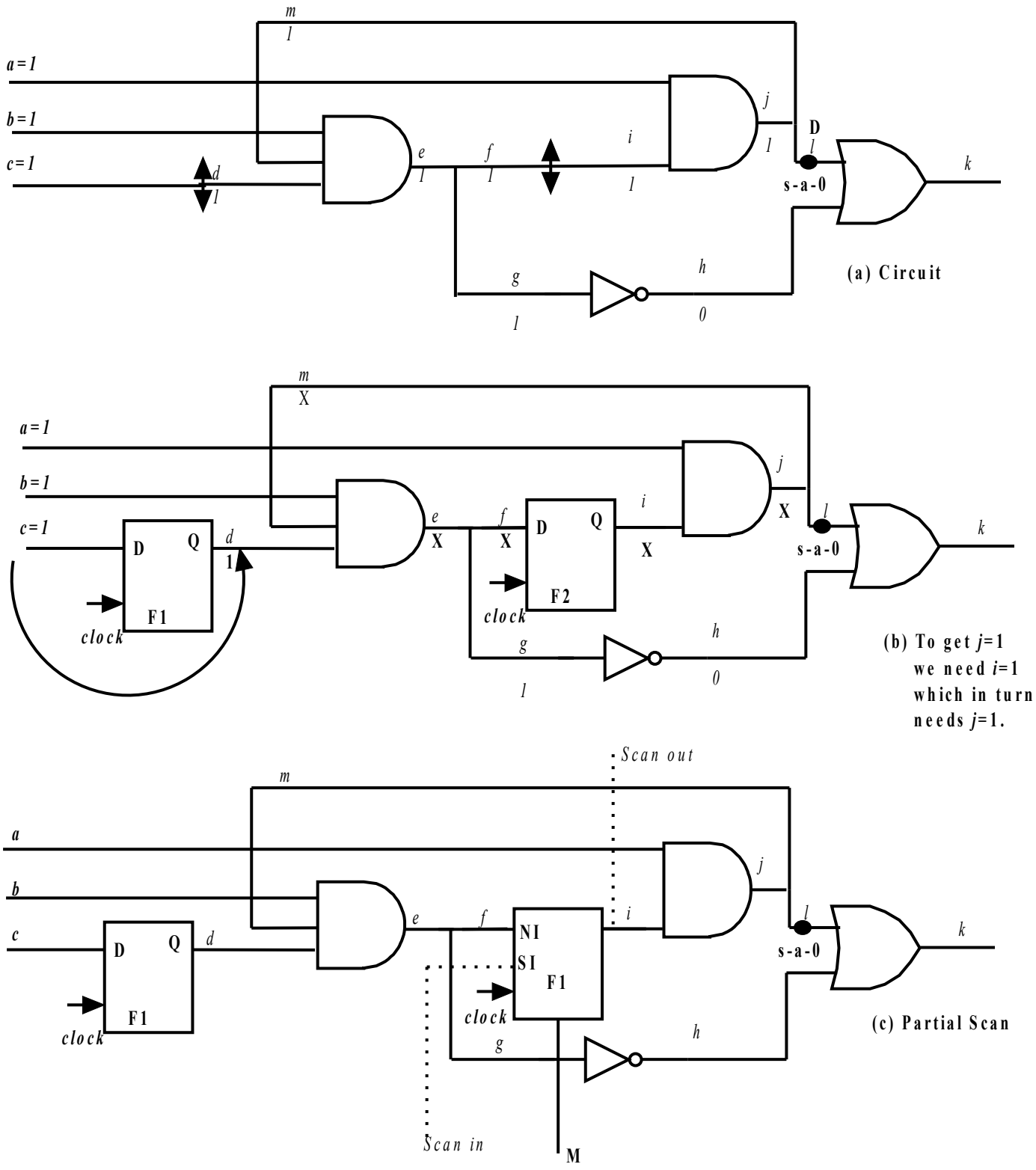


Figure 18. Cyclic circuits and requirement of scan chain

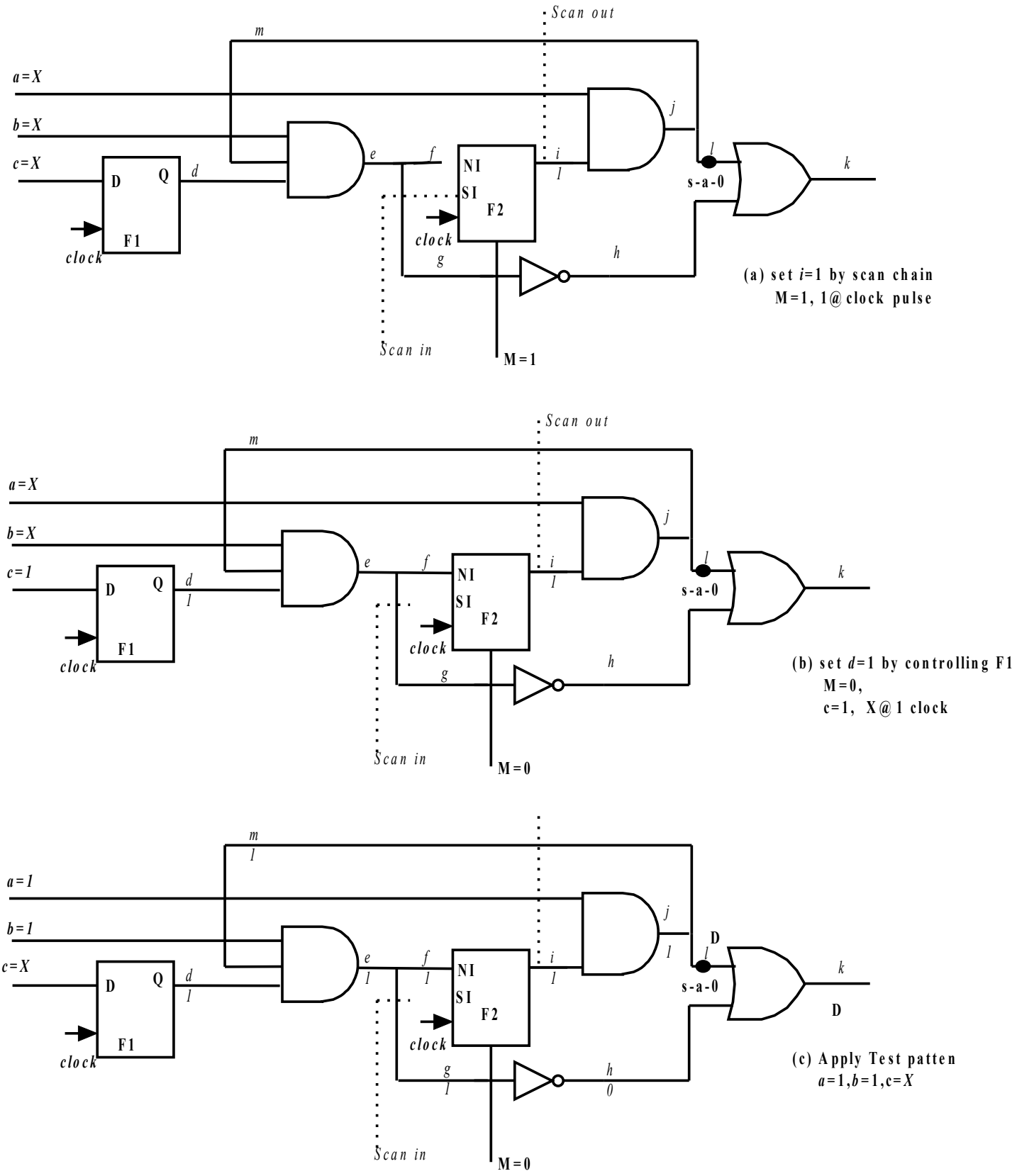


Figure 19. ATPG and testing using partial scan technique

The procedure of testing the circuit of Figure 18(c) using partial scan is shown in Figure 19. As shown in Figure 19(a), net *i* is set to 1 by making M=1 and applying 1 in Scan in input with a clock pulse; F2 is set using scan chain. Now (Figure 19(b)) net *d* is set to 1 by making M=0 and applying 1 in primary input *c* with a clock pulse; F1 is set using time frame expansion. Finally after setting the flip-flops, test pattern *a*=1,*b*=1,*c*=X is applied, which sensitizes and propagates the fault effect to the output *k*.

6. Conclusions

In the combined two lectures we have seen testing of sequential circuits with design for testability techniques, namely set/reset lines, shift register with set/reset lines, scan chain and partial scan chain. Design for testability is additional resources like circuitry or I/O pins used to make testing easier. Among all the techniques, we saw that partial scan is the most efficient one as it uses additional resources only when testing becomes infeasible due to cyclic parts in a circuit. However, sometimes non-cyclic flip-flops are also made scan enabled (in partial scan technique) to minimize ATPG time; as shown in Figure 17 of this lecture, propagating fault effects by scan chains minimize ATPG effort. Another direction called "multiple chains" is taken to minimize test time in scan chain based techniques. In this scheme, instead of a single scan chain, the chain is broken into multiple parts. The start of each chain is brought out as separate Scan in and end of the chain is brought out as separate Scan out. This increases the number of I/Os, but all the smaller chains can be fed with bits values concurrently, minimizing test time.

References

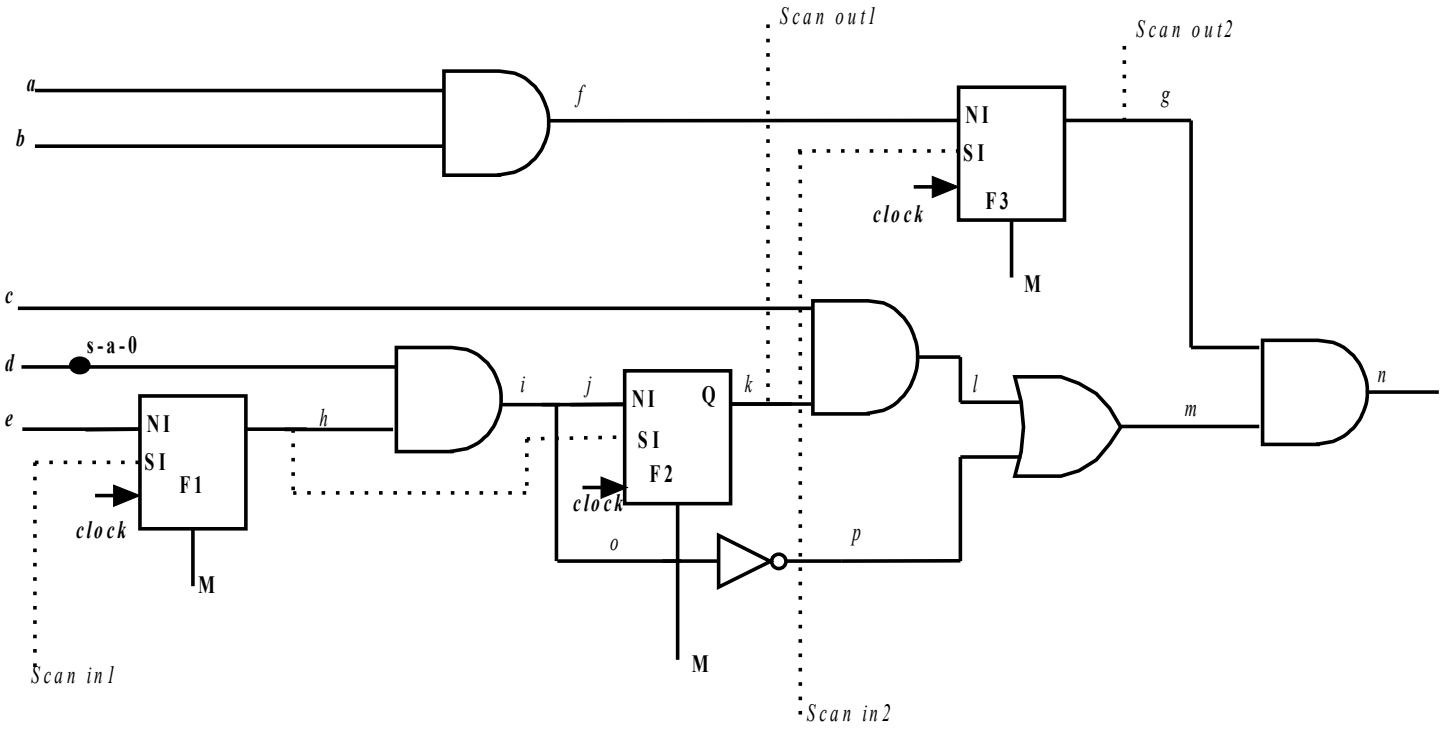
1. M. J. Y. Willaims and J. B. Angell, "Enhancing Testability of Large-Scale Integrated Circuits via Test Points and Additional Logic," IEEE Trans. on Computers, vol. C-22, no. 1, pp. 46–60, 1973.
2. K.-T. Cheng and V. D. Agrawal, "A Partial Scan Method for Sequential Circuits with Feedback," IEEE Trans. on Computers, vol. 39, no. 4, pp. 544–548, Apr. 1990.

Questions and Answers

1. How can test time be reduced for the circuit with scan chain of Figure 14 of this lecture?

   Answer:

   The circuit in Figure 14 has a single scan chain with three flip-flops. So time taken to set/reset all the flip-flops is three clock pulses. To save test time the scan chain can be divided into multiple sub-chains with separate Scan in and Scan out pins. In this case we have divided the chain into 2 parts—one has F1 and F2 and the second has F3 only. So we have two sets of Scan in and Scan out pins. The design is illustrated in the figure below. Now both the chains can be loaded concurrently. So, only two clock pulses are required to set/reset all the flip-flops.



2. How can the flip-flops themselves be tested in a scan chain based testing?

Answer:

An advantage of scan chain based testing over time frame expansion and shift register technique is the ability to test the flip-flops in the circuit under test. A toggle sequence, 00110011 . . ., of length $n_{ff} + 4$, where n_{ff} is the total number of flip-flops, is applied at Scan in. This sequence produces all four transitions $0 \rightarrow 0$, $0 \rightarrow 1$, $1 \rightarrow 1$ and $1 \rightarrow 1$ in all the flip-flops. The sequence is brought out through the Scan out pin. So if a correct toggle sequence is observed in the Scan out pin, then the flip-flops are working normally. It is shown in stuck at fault model that toggle sequence covers almost all stuck at faults in the flip-flops.